

Final Project Paper

Material Reduction of Pocket Carabiner

ME424

Dustin Leung

Professor Bay

1. Abstract

Carabiners are used for various applications, and in this project, we are focusing on those designed to hold small bags. The goal is to optimize the geometry of a pocket carabiner to reduce material usage while maintaining similar structural integrity. Using finite element analysis, different variations based on the standard carabiner design will be tested to evaluate how removing material from specific areas affects its strength. Multiple iterations will be analyzed, and we will develop a method to determine which variation is most successful in reducing weight while preserving strength. Our analysis will examine how removing mass from the carabiner's spine minimally impacts the distribution of stress in the part, as well as how excessive material removal significantly weakens the structure. This project will demonstrate how a pocket carabiner can be improved by reducing material usage, thereby lowering production costs and increasing efficiency.

2. Background

2.1. Context and Motivation

While considering potential projects, I looked at my wallet and noticed the carabiner I use to connect it to my keys. This carabiner detaches from my wallet and hangs on the wall. To my knowledge, it should be strong enough to hold small bags, and at the very least, a backpack. However, upon closer inspection, I felt there might be a way to make it lighter and more optimized. Based on my understanding of bending stress in the tube, some material could potentially be removed. My prediction is that certain edges of the spine experience primarily axial stress, which is generally much smaller compared to bending stress.

2.2. Selected Part

The overall design of the part is based on a basic D-shaped carabiner, with one side featuring a tubular spine and the other side incorporating a simple hooking mechanism that is pushed in to attach to a hook. The carabiner is approximately 45–50 mm in length, making it much smaller than those used for activities like rock climbing, which are designed to support significantly greater loads. As shown in Figure 1, the carabiner's general shape is illustrated. The part is assumed to be made of Aluminum 7075 alloy, a standard material for carabiners. The model was created in NX and imported into Ansys for finite element analysis.



Fig. 1: Image of Carabiner

In figure 2, you can see the baseline component, and where the loading conditions are going to be applied.

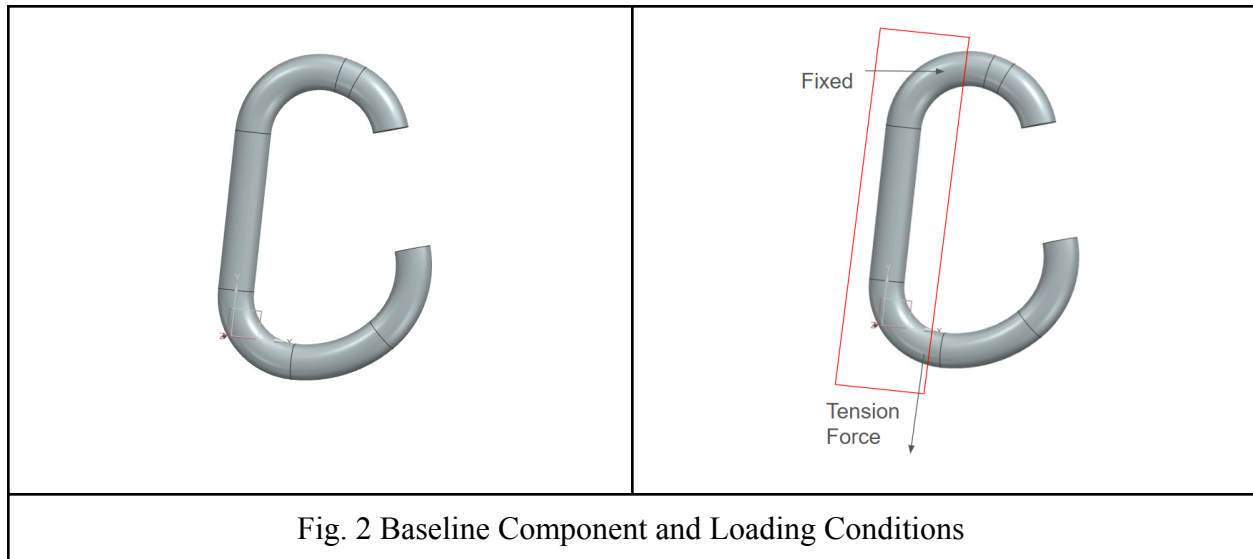


Fig. 2 Baseline Component and Loading Conditions

2.3. Objectives and Constraints

The objective of the design modification is to reduce the weight of the carabiner while maintaining its general stress and deflection characteristics. This will be achieved by decreasing the volume of the design, as volume directly correlates with weight since all variations are made from the same material. The goal is to create a more cost-effective product while preserving its intended use cases. Additionally, the overall size constraints must remain unchanged, as the fundamental design of the carabiner should not be altered. This optimization is intended for everyday use rather than specialized applications.

2.4. Variations

The design variations I will test involve modifying certain sections of the carabiner from a tubular shape to a rectangular shape. Based on my assumptions, some areas of the spine experience minimal stress, and removing material from these regions should not significantly affect the overall strength of the carabiner. For consistency, all tests will use the same loading condition: a downward force of 360 N (approximately 80 pounds). This represents the upper limit of the weight this carabiner is intended to support, as its primary function is to hang items like bags. Small carabiners like this are not designed for climbing or supporting body weight, and product descriptions for similar carabiners often explicitly state that they are not suitable for climbing purposes.

3. Methods

3.1. Simplified Model

The simplified model used for mesh convergence analysis is the same one shown in Figure 2. This model represents one way to conceptualize a D-shaped carabiner, as it is designed to minimize stress on the connection side and instead concentrate most of the stress on the spine. To better understand this, I removed the connection piece when creating my model. This approach aligns with the methodology used in my Homework 5 FEA analysis, where a referenced paper employed the same simplification. The D-shape was chosen because it matches the design of my project inspiration—the pocket carabiner that holds my keys.

3.2. Mesh Convergence and Validation

To achieve mesh convergence, we assumed that curvature capture would be used due to the tubular shape of the model. To determine when the mesh converges, we progressively reduced the element size and observed when

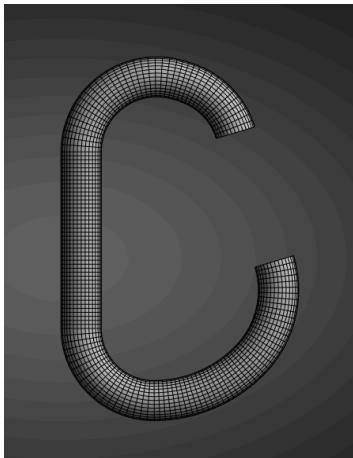
the maximum von Mises stress plateaued. This indicates that further refinement does not significantly change the results, signifying convergence. The following table presents the results of my mesh analysis.

Element Size	Von-Mises Equivalent Stress
1mm	995 MPa
0.75mm	974 MPa
0.5mm	971 MPa
0.25mm	970 MPa

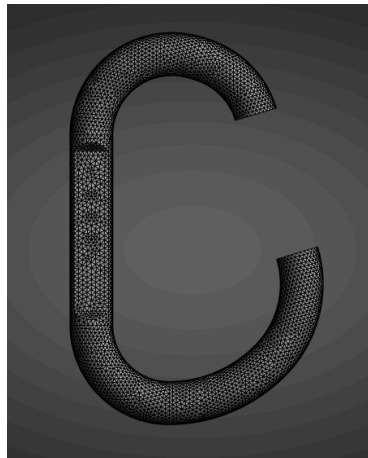
As the element size decreases, the equivalent stress also decreases, which aligns with expectations, as the FEA analysis becomes more accurate. When refining the mesh from 0.5 mm to 0.25 mm element sizes, the change in stress values was minimal. However, from my experience, the computation time increased dramatically—from a few seconds to several minutes—as the software struggled to generate a mesh with such small elements. Thus, I concluded that the mesh converges at a 0.5 mm element size with curvature capture included. This mesh convergence analysis was based on an initial loading condition that may not accurately reflect the typical forces acting on a small carabiner. Small carabiners are not designed to support large, heavy objects, and under the tested load, the equivalent stresses indicated failure. Initially, I aimed for a worst-case scenario of 1.5 times my body weight, but further research revealed that this type of carabiner is explicitly not intended to support human weight. Nonetheless, even with a load leading to failure, the mesh convergence still occurred, meaning the results remain valid for analysis.

3.3. Final Mesh with Loading Conditions

The final resulting mesh is shown in figure 3. It uses the converged mesh calculated above.



Standard

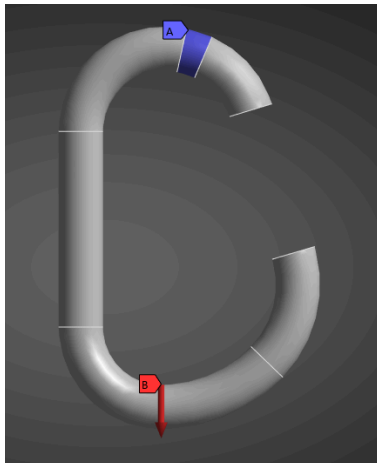


Reduced Spine

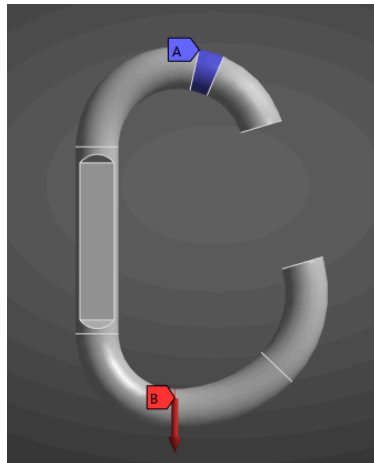


Whole Tube Reduced

Fig. 3 Final Meshes used for Analysis



Standard



Reduced Spine



Whole Tube Reduced

Fig. 4 Loading Conditions on each Variation

In Figure 4, the blue-labeled point **A** represents the fixed part of the model, while the red marker labeled **B** indicates the location of the applied force. As previously stated, the applied force is 360 N, or approximately 80 pounds.

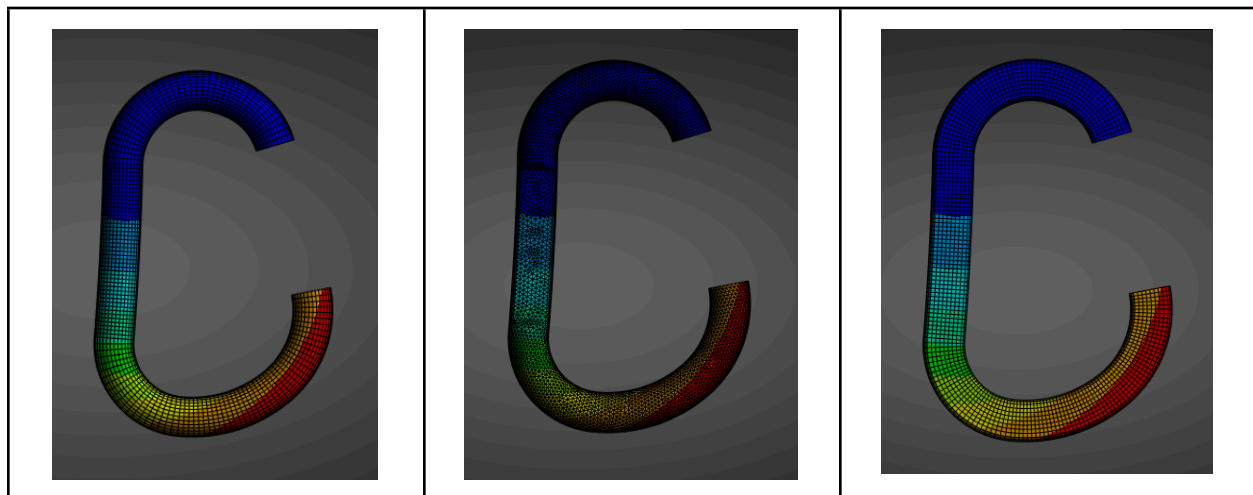
3.4. Material Properties

The material assumed for the carabiner in this analysis is Aluminum 7075. The material properties were obtained from MatWeb to conduct the FEA analysis. Aluminum 7075 was chosen because I was unable to determine the exact metal used in my pocket carabiner. However, after researching common materials for carabiners, Aluminum 7075 appeared to be the more frequently used option.

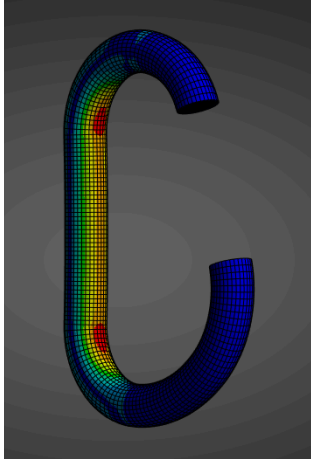
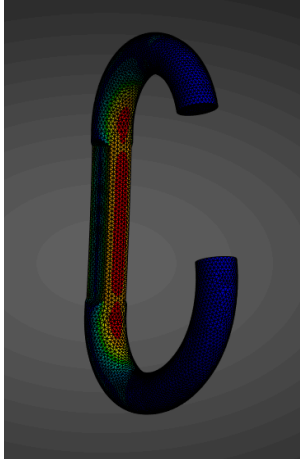
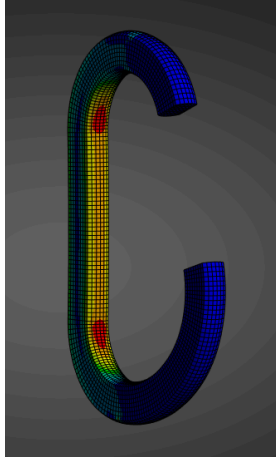
Material	Young Modulus	Poisson's Ratio	Tensile Yield Strength
Aluminum 7075	71.7 GPa	0.33	503 MPa

4. Results

4.1. Deformation and Stress



Standard	Reduced Spine	Whole Tube Reduced
Fig. 5 Deflection / Deformed Model Shape		

		
Standard	Reduced Spine	Whole Tube Reduced
Fig. 6 Points of High Stress (von-Mises)		

From these images, it is clear that the deflection characteristics remain relatively constant across the variations. Additionally, it can be observed that as the spine is reduced, the stress along the spine increases and becomes more concentrated at the points where the bends begin.

4.2. Quantitative Results

Variation	Maximum Deflection (mm)
Standard	1.63
Reduced Spine	1.79
Whole Tube Reduced	1.94

Variation	Maximum Stress (von-Mises) (MPa)
Standard	349.82
Reduced Spine	358.76
Whole Tube Reduced	416.18

Variation	Volume (mm ³)
Standard	1734.4
Reduced Spine	1627.8
Whole Tube Reduced	1251.5

Variation	Maximum Stress Change / Volume Reduction	Result
Standard	$(349.82-349.82)/(1734.4-1734.4)$	0 (not relevant)
Reduced Spine	$(358.76-349.82)/(1734.4-1627.8)$	0.0838
Whole Tube Reduced	$(416.18-349.82)/(1734.4-1251.5)$	0.1374

The first result isn't important, as we are trying to do everything relative to the standard variation, which is what I am trying to improve.

5. Discussion and Conclusions

5.1. Discussion & Results Interpretation

The results show that as more weight is reduced, the maximum deflection increases, which aligns with expectations. With less cross-sectional area, there is less resistance to deflection. The increase in deflection—about 0.31 mm between the smallest and largest maximum deflections—is not catastrophic. We can expect the carabiner to maintain its general shape across all variations, making all three options viable.

The more notable change is in the maximum von Mises stress. When mass is only removed from the spine, the maximum stress change increases by around 9 MPa. However, when more mass is removed from the entire design, the maximum stress change rises to about 65 MPa, which is a significant difference. Despite this large difference in stress, the carabiner does not fail, as the stress remains below the tensile yield strength of 503 MPa for Aluminum 7075. To better interpret the results, we created a scoring system based on how the reduction in volume, which directly translates to mass and weight, affects the maximum stress change. Since both designs technically meet the loading conditions we tested, we needed a way to quantify which one is more efficient. We decided to assess how much the maximum stress changed per unit of volume reduced. This approach allows us to identify which variation most effectively reduces volume while maintaining strength.

Using our new scoring system, we find that the reduced spine design offers the most efficient material reduction while retaining more of its strength properties. Logically, this interpretation makes sense. The highest stress locations for all three designs are at the bend where the straight spine transitions to the hooking parts, which is expected as this is where most of the bending stress will be concentrated. When only removing material from the spine, we observe that while the stress in this area increases, it does not significantly alter the location of the maximum stress. Most of the increased stress is concentrated along the spine, raising its stress concentration, but not beyond the stress concentration at the curve, meaning the part will not fail at the spine. On the other hand, removing volume from the entire part leads to a large increase in maximum stress at the curves, as the cross-sectional area is reduced. This aligns with our expectations, as the curves experience greater stress when the material is reduced. Overall, the findings from our variations make logical sense.

For a cheaper and only marginally weaker small carabiner, reducing material from the spine will maintain very similar strength properties across the entire carabiner, while also lowering production costs. Given that this is a

mass-produced product, any reduction in cost improves profit margins, making it a valuable improvement for the company.

5.2. Future Work

For future work, one aspect I disregarded in my analysis was the impact of the connection piece on deflection and stress in carabiners. In my research for this project, the FEA paper I reviewed treated the connection piece as open, and the general carabiner design directs most of the stress through the spine. However, in terms of deflection, the part that experiences the most deflection is still connected to the fixed part at the top, meaning the connection piece itself would need to deflect significantly. This may not reflect how the carabiner behaves in reality, leading me to believe that if I were to conduct future research, I would investigate how the connection piece interacts with standard loading conditions.

5.3. Conclusions

- Among the three tested variations, the most efficient material reduction is in the spine.
- Removing material from the entire tube significantly reduces mass but greatly increases maximum stress, bringing the structure closer to failure.
- Material reduction in the spine maintains similar maximum stress concentrations and structural stability while lowering costs.

References

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